Mavs Chung

332 formula.

M_1

target Pop. = $U = \{1, 2, \cdots N\}$ complete population.

sampled Pop. = collection that could be included by sampling design.

Sampling Unit : object being sampled.

sampling frame : actual list of unit that could be in sample.

Observation Unit : object we take measurements.

Errors :

i. frame construction : coverage error

under / over coverage

ii. Obtaining Sample : (sampling design)

focus → Probability SD : probabilistic method (sampling error)

Non-probability SD : based on subjective criteria

⇒ non-response error & selection bias.

iii. Measurement Error : trivial

Prob-based SD : $U = \{1, 2, \ldots N\}$ Pop. Values.

$\rightarrow y_1 \; y_2 \quad y_n$

$\mu = \frac{1}{N} \sum_{i \in U} y_i$

$\tau = \sum y_i$

$\sigma^2 = \frac{1}{N-1} \sum (y_i - \mu)^2 = \frac{1}{N-1} \left(\sum y_i^2 - N\mu^2 \right)$

$S \Rightarrow$ r.v.

$s \Rightarrow$ realized sample.

take subset S and estimate quantiles. add $\hat{}$ and $\sum_{i \in S}$

Inclusion Prob: $\pi_i = P(i \in S)$, $\pi_{ij} = P(i, j \in S)$

Estimators = same formula w/ $\sim$

(random based on S)

* SRSWOR: take sample of size n out of N.

, all possible size n are equally likely.

total possible sample size $n = \binom{N}{n}$

$\Rightarrow P(S) = \frac{1}{\binom{N}{n}}$

$\Rightarrow \pi_i = \frac{n}{N}$, $\pi_{ij} = \frac{n(n-1)}{N(n-1)}$

☆ $\begin{cases} \hookrightarrow E(\tilde{\mu}) = \mu \\ Var(\tilde{\mu}) = \left(1-\frac{n}{N}\right)\frac{\sigma^2}{n} \end{cases}$ ← pop. Var. (unknown)

for $\sqrt{Var(\tilde{\mu})}$

for CI you use sample Var $\hat{\sigma}^2$ since unbiased

$\Rightarrow SE(\hat{\mu}) = \sqrt{\left(1-\frac{n}{N}\right)\frac{\hat{\sigma}^2}{n}}$

b/c calculated from "sampled" $\hat{\sigma}^2$

$\tilde{\mu}$ in terms of indicator I_i $\begin{cases} 1, & i \in S \\ 0 & o/w \end{cases}$

$\Rightarrow \hat{\mu} = \frac{1}{n}\sum_{i \in S} y_i = \frac{1}{n}\sum_{i=1}^{N} I_i y_i$

$Bias(\tilde{\theta}) = E(\tilde{\theta}) - \theta$

* 95% CI = $\tilde{\theta} \pm z_{\alpha/2}\, SE(\hat{\theta})$

95% ⇒ 0.975 = 1.96*

τ in SRS $= \sum_{i \in U} y_i = N\mu$

$\Rightarrow \hat{\tau} = N\hat{\mu} \quad (\hat{\theta})$

$Var(\tilde{\tau}) = Var(N\hat{\mu}) = N Var(\tilde{\mu})$

Proportion Value:

$\sigma^2 = \frac{N}{N-1} p(1-p)$

FPC (finite population correction)

- $\frac{n}{N}$ = sampling fraction
- $(1-\frac{n}{N})$ = FPC

if $n \approx N$, $Var(\tilde{\mu}) = 0$, sampling Variability gone.

$N >> n$, FPC ≈ 1

gaurantee CI width $< C$

let CI $< C$ and $\Rightarrow n > \ldots$

Ratio est.

Measure 2 attribute: $(x_1, y_1), (x_2, y_2) \ldots$

$\mu_x = \frac{1}{N} \sum_{i \in U} x_i$, $\mu_y = \frac{1}{N} \sum_{i \in U} y_i$

$\hat{\rho} = \sum_{i \in U} \frac{(x_i - \hat{\mu}_x)(y_i - \hat{\mu}_y)}{(N-1)\, \hat{\sigma}_x \hat{\sigma}_y}$ (for pop., no $\wedge$)

Ratio $\hat{\theta} = \frac{\hat{\mu}_y}{\mu_x} = \frac{\tau_y}{\tau_x}$

if μ_x known, estimate $\hat{\mu}_y$ by $\hat{\mu}_y \cdot \frac{\mu_x}{\hat{\mu}_x} = \hat{\theta} \cdot \mu_x$

side $Var(\tilde{\theta}) = \frac{1}{\mu_x^2} (1-\frac{n}{N}) \frac{\sigma_r^2}{n}$ residual Var.

Ⓡ svyratio (~y , ~x , design) $\begin{cases} \hat{\theta} \\ SE(\hat{\theta}) \\ \widehat{Var}(\hat{\theta}) \end{cases}$

Ratio ester

$E(\tilde{\theta}) \approx \frac{\mu_y}{\mu_x} = \hat{\theta}$

$Var(\tilde{\theta}) = \frac{1}{\mu_x^2}(1-\frac{n}{N})(\frac{1}{n})(\sigma_y^2 + \theta^2\sigma_x^2 - 2\theta\rho\sigma_x\sigma_y)$

$SD(\hat{\theta}) = \sqrt{\frac{1}{\mu_x^2}(1-\frac{n}{N})(\frac{1}{n})(\hat{\sigma}_y^2 + \hat{\theta}^2\hat{\sigma}_x^2 - 2\hat{\theta}\hat{\rho}\hat{\sigma}_x\hat{\sigma}_y)}$

$\tilde{\theta}$ work well when $Var(\hat{\theta})$ small.

Ideally, $\rho \approx 1$

$y_i \approx \theta x_i \Rightarrow \sigma_r^2$ small $\Rightarrow \tilde{\theta}$ accurate.

If μ_x known estimate μ_y with

$\hat{\mu}_{ratio} = \hat{\theta}\mu_x$

$\widehat{Var}(\hat{\mu}_{ratio}) \approx (1-\frac{n}{N})(\frac{1}{n})(\hat{\sigma}_y^2 + \hat{\theta}^2\hat{\sigma}_x^2 - 2\hat{\theta}\hat{\rho}\hat{\sigma}_x\hat{\sigma}_y)$

regression est.

$\tilde{\mu}_{*}^{reg} = \hat{\mu}_y + b(\mu_x - \tilde{\mu}_x)$ svyglm(~y, ~x, ...)

$E(\tilde{\mu}_*) = \mu_y$

$Var(\hat{\mu}_*) = (1-\frac{n}{N})(\frac{1}{n})(\sigma_y^2 + b^2\sigma_x^2 - 2b\rho\sigma_x\sigma_y)$

minimize Var $\Rightarrow b = \frac{\hat{\rho}\hat{\sigma}_y}{\hat{\sigma}_x}$

$\Rightarrow \widehat{Var}(\hat{\mu}_{reg}) = \underbrace{(1-\frac{n}{N})(\frac{1}{n})}_{Var(\hat{\mu}_y)}(1-\hat{\rho}^2)\hat{\sigma}_y^2$

<u>$\hat{\mu}_{ratio}$</u> better than <u>$\hat{\mu}_y$</u> when

$\hat{\theta}^2\hat{\sigma}_x^2 - 2\hat{\theta}\hat{\rho}\hat{\sigma}_x\hat{\sigma}_y < 0 \Leftrightarrow \hat{\rho} > \frac{\hat{\theta}\hat{\sigma}_x}{2\hat{\sigma}_y}$

∝ mean/Var similar for x, y and $\hat{\rho} > 0.5$

<u>$\hat{\mu}_{reg}$</u> better than <u>$\hat{\mu}_y$</u> when $\hat{\rho} \neq 0$

divide N to H strata

$N = \sum_{h=1}^{H} N_h$ ← pop size of stratum h

Stratum	Unit	Measurements / Values
1	$\{1, 2, \dots N_1\}$	$y_{11}, y_{12} \dots y_{1N_1}$
2	$\{1, 2, \dots N_2\}$	$y_{21}, y_{22} \dots y_{2N_2}$
$\vdots$	$\vdots$	$\vdots$
H	$\{1, 2 \dots N_H\}$	$y_{H1}, y_{H2} \dots y_{HN_H}$

take SRS of size n_h from N_h in $h = 1, 2, \dots H$

pop quantiles : μ = all pop. mean.

μ_h = pop mean of stratum h.

σ_h^2 = pop variance of ...

sample quantiles : $\hat{\mu}_h$ = sample mean in stratum h

$\hat{\sigma}_h^2$ = sample variance in ...

Strata

- $\mu = \sum_{h=1}^{H} \omega_h \mu_h \Rightarrow \hat{\mu}_S = \sum_{h=1}^{H} \omega_h \hat{\mu}_h$ (V) ← $\omega_h = \frac{N_h}{N}$
- fact : more samples in SRS.
- $E(\hat{\mu}_S) = \mu$
- $Var(\hat{\mu}_S) = \sum_{h=1}^{H} \omega_h^2 (1 - \frac{n}{N}) \frac{\sigma_h^2}{n_h}$

Strata alloc.

Allocation approach 1 : proportional allocation

$\Rightarrow n_h = n \frac{N_h}{N} = n W_h$

total sample size

$Var(\hat{\mu}_{s,prop}) = \sum_{h=1}^{H} W_h \left(1 - \frac{n}{N}\right) \frac{\sigma_h^2}{n}$

stratify > SRS when $\sum_{h=1}^{H} W_h \sigma_h^2 < \sigma^2$

$\hookrightarrow \sigma^2 = \sum_{h=1}^{H} W_h \sigma_h^2 + \sum_{h=1}^{H} W_h (\mu_h - \mu)^2$

Allocation approach 2 : optimal allocation (min Var)

$\Rightarrow n_h = \frac{W_h \sigma_h}{\sum_{h=1}^{H} W_h \sigma_h} \cdot n \propto W_h \sigma_h$

$\hookrightarrow$ sample more from large N_h & large σ_h

when σ_h all same $\rightarrow$ proportional also optimal.

$t\{\alpha/2, df\} \rightarrow$ pop Var σ^2 is known

$z\{\alpha/2, df\} \rightarrow$ pop SE σ is known.

M2

psu = cluster

ssu = unit within cluster

A clusters w/ B units each, $\Rightarrow N = AB$

↳ take SRS #a clusters and # b unit each.

Sample size $n = ab$

Case ① $b = B$

pop. $\mu = \frac{1}{A} \sum_{i=1}^{A} \mu_i$,

$\hat{\mu}_{clus} = \frac{1}{a} \sum_{i \in S} \mu_i$

$\widehat{Var}(\tilde{\mu}_{clus}) = (1 - \frac{a}{A}) \frac{\hat{\sigma}_A^2}{a}$

↳ $\sigma_A^2 = \frac{1}{A-1} \sum_{i=1}^{A} (\mu_i - \hat{\mu})^2$ pop Var of cluster

(SRS) $\frac{n}{N} = \frac{aB}{AB} = \frac{a}{A}$ (cluster)

$\frac{Var(\tilde{\mu}_{clus})}{Var(\hat{\mu})} = B \cdot \frac{\sigma_A^2}{\sigma^2}$ $\frac{\rightarrow \text{clusters } \Sigma A}{\rightarrow \text{all unit } \Sigma A \Sigma B}$

cluster works poor if cluster similar

$\sigma_B^2 = \frac{1}{A} \sum_{a=1}^{A} \left(\frac{1}{B-1} \sum_{b=1}^{B} (y_{ab} - \mu_a)^2 \right)$

$\sigma^2 = \frac{1}{AB-1} (\sigma_b^2 \cdot A(b-1) + \sigma_A^2 \cdot B(A-1))$

AB large $\Rightarrow \sigma^2 \approx \sigma_A^2 + \sigma_B^2$

Case ② $n = ab$

$$\hat{\mu}_{clus,2} = \frac{1}{a}\sum_{i \in S} \mu_i = \frac{1}{a}\sum^{a}\left(\frac{1}{b}\sum^{b} y_{ij}\right)$$

$$\widehat{Var}(\hat{\mu}_{clus,2}) = \left(1-\frac{a}{A}\right)\frac{\hat{\sigma}_A^2}{a} + \left(1-\frac{b}{B}\right)\frac{\hat{\sigma}_B^2}{ab}$$

$$\hat{\tau} = N\mu_{clus} = AB\mu_{clus}$$

factor : controlled input (type of drink)

factor levels : possible values of factor (coffee vs. tea)

treatment : combination of factors (coffee + chocolate)

experimental unit : obj recieving treatment (student)

response : measured outcome (cts, normal)

replication : reduce variance

randomization : remove bias

blocking : increase precision.

" block what you can, randomise what you can't "

Single

Single mean/treatment : $Y_i = \mu + \epsilon_i$

$\epsilon_i \sim N(0, \sigma^2)$, iid.

$\hat{\mu} = \bar{y}$, $SE(\hat{\mu}) = \frac{\hat{\sigma}}{\sqrt{n}}$

$H_0 : \mu = \mu_0$, $H_A : \mu \neq \mu_0$

test

$\frac{\hat{\mu} - \mu_0}{\frac{\hat{\sigma}}{\sqrt{n}}} = t \Rightarrow$ reject if $|t| > c$

$\alpha = 0.05$

$\Rightarrow P(|T| \leq c) = 0.95$

$\Rightarrow$ df & 0.975.

Two

Two treatment: $Y_{ij} = \mu + \tau_i + \epsilon_{ij}$

μ = overall mean , $\tau_{1,2}$ = treatment effects

$\epsilon_{ij} \sim N(0, \sigma^2)$ iid. same σ^2 for both treatment.

$n_{1,2}$ = units in treatment 1 or 2

$n = n_1 + n_2$

$i = 1, 2$, $j = 1, 2, \dots n_i$

. = average over that index.

$\bar{Y}_{1.} = \frac{1}{n_1} \sum_{j=1}^{n_1} Y_{1j}$ $\bar{Y}_{2.} = \frac{1}{n_2} \sum_{j=1}^{n_2} Y_{2j}$

$\bar{Y}_{..} = \frac{1}{n_1 + n_2} \sum_{i=1}^{2} \sum_{j=1}^{n_i} Y_{ij}$

$E(\bar{Y}_{1.}) = \mu + \tau_1$, $E(\bar{Y}_{2.}) = \mu + \tau_2$

$E(\bar{Y}_{..}) = \mu$

$\hat{\mu} = \bar{y}_{..}$, $\hat{\tau}_1 = \bar{y}_{1.} - \bar{y}_{..}$, $\hat{\tau}_2 = \bar{y}_{2.} - \bar{y}_{..}$

$$\hat{\sigma}^2 = \frac{(n_1 - 1)\hat{\sigma}_1^2 + (n_2 - 1)\hat{\sigma}_2^2}{n_1 + n_2 - 2}$$

$H_0 : \tau_1 = \tau_2$

$$t = \frac{\hat{\tau}_1 - \hat{\tau}_2}{\hat{\sigma}\sqrt{\frac{1}{n_1} + \frac{1}{n_1}}}$$ reject $t > C$. df $= n_1 + n_2 - 2$

tapply(dat\$A , dat\$B , mean)

→ 1 mean per group B

Two treatment w/ block (single replicate)

$Y_{ij} = \mu + \tau_i + \beta_j + \epsilon_{ij}$.

$i = 1, 2$, $j = 1, 2 \dots b$

constraint : $\sum \tau_i = 0$ & $\sum \beta_j = 0$

$\hat{\mu} = \bar{y}_{..}$

$\hat{\tau}_i = \bar{y}_{i.} - \bar{y}_{..}$ $\quad \hat{\sigma}^2 = \frac{1}{b-1} \sum_{i=1}^{2} \sum_{j=1}^{b} (y_{ij} - \mu - \tau_i - \beta_j)^2$

$\hat{\beta}_j = \bar{y}_{.j} - \bar{y}_{..}$

$H_0 = \tau_1 = \tau_2$, $t = \frac{\hat{\tau}_1 - \hat{\tau}_2}{\hat{\sigma}\sqrt{\frac{2}{b}}} \sim t_{b-1}$

CRD (>2 treatments)

t treatments w/ r replicates . $\Rightarrow n = tr$

$Y_{ij} = \mu + \tau_i + \epsilon_{ij}$, $i = 1 \dots t$, $j = 1 \dots r$.

$\mu + \tau_i$ = mean response for trt i.

$\hat{\mu} = \bar{y}_{..}$, $\hat{\tau}_i = \bar{y}_{i.} - \bar{y}_{..}$

$\hat{\tau}_i - \hat{\tau}_j = \hat{y}_{i.} - \hat{y}_{j.}$ $\quad df = t(r-1)$

$\hat{\sigma}^2 = \frac{1}{t(r-1)} \sum_{i=1}^{t} \sum_{j=1}^{r} (y_{ij} - \hat{\mu} - \hat{\tau}_i)^2$

α_i = main effect of level i of factor A

Contrast $\theta = \sum_{i=1}^{t} a_i \tau_i$ s.t. $\sum_{i=1}^{t} a_i = 0$

$\hat{\theta} = \sum_{i=1}^{t} a_i \bar{y}_{i.}$, $E(\hat{\theta}) = \theta$ $\qquad \frac{\tilde{\theta} - \theta}{\tilde{\sigma}\sqrt{\frac{\sum a_i^2}{r}}} \sim t_{t(r-1)}$

$Var(\hat{\theta}) = \frac{\hat{\sigma}^2}{r} \sum_{i=1}^{t} a_i^2$

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$H_0: \tau_i = 0$, $H_A\ \tau_i \neq 0$

If Ho true, $MS = \tilde{\sigma}_t^2 = \frac{r}{t-1} \sum_{i=1}^{t} (\bar{Y}_{i.} - \bar{Y}_{..})^2$

$$\frac{\tilde{\sigma}_t^2}{\tilde{\sigma}^2} \sim F_{t-1,\, t(r-1)} \quad \left(\frac{MS\ \text{treatment}}{MS\ \text{error}}\right)$$

If Ho false

$$E(\tilde{\sigma}_t^2) = \sigma^2 + \frac{r}{t-1} \sum_{i=1}^{t} \tau_i^2$$

reject Ho if

1. $F >> 1$
2. $F > C$, $P(F_{t-1,\, t(r-1)} \geq c) = \alpha$ n. dn.

Variance of entire sample : $\frac{1}{tr-1} \sum_{i=1}^{t} \sum_{j=1}^{r} (y_{ij} - \bar{y}_{..})^2$

estimate residual variance $\hat{\sigma}^2 = \frac{1}{t(r-1)} \sum_{i=1}^{t} \sum_{j=1}^{r}$

$$\hat{\sigma}_t^2 = \frac{r}{t-1} \sum_{i=1}^{t} (\bar{y}_{i.} - \bar{y}_{i i})^2$$

TSS = trtSS + RSS (or SSE)

Anova (1-factor , t trts. , r replicates)

Source	SS	df	MS	F
trt	$r \sum_i (\bar{y}_{i.} - \bar{y}_{..})^2$	$t-1$	$\hat{\sigma}_t^2$	$\frac{\hat{\sigma}_t^2}{\hat{\sigma}}$
Residual	$\sum_i \sum_j (y_{ij} - \bar{y}_j)^2$	$t(r-1)$	$\hat{\sigma}^2$	
total	$\sum_i \sum_j (y_{ij} - \bar{y}_{..})^2$	$tr-1$		

Mz

CRD (# replicates differs)

$Y_{ij} = \mu + \tau_i + \epsilon_{ij}$ (same as CRD)

r_i = replicates in trt i

trt mean : $\bar{Y}_{i\cdot} = \frac{1}{r_i}\sum_{j=1}^{r_i} Y_{ij}$

grand mean : $\bar{Y}_{\cdot\cdot} = \frac{1}{N}\sum_{i=1}^{t}\sum_{j=1}^{r_i} Y_{ij}$ where $N = \sum_{i=1}^{t} r_i$

$E(\bar{Y}_{\cdot\cdot}) = \mu$

$\hat{\mu} = \bar{Y}_{\cdot\cdot}$

residual Var.

$\hat{\sigma}^2 = \frac{1}{\sum_{i=1}^{t} r_i - t}\sum_{i=1}^{t}\sum_{j=1}^{r_i}(Y_{ij} - \hat{\mu} - \hat{\tau}_i)^2$

$\hat{\tau}_i = \bar{Y}_{i\cdot} - \bar{Y}_{\cdot\cdot}$

Contrast : $\theta = \sum_{i=1}^{t} a_i \tau_i \Rightarrow \theta = \sum_{i=1}^{t} a_i \bar{Y}_{i\cdot}$

$\hat{\theta} \sim N(\theta, \sigma^2 \sum_{i=1}^{t} \frac{a_i^2}{r_i})$

$H_0 : \theta = 0 \Rightarrow t_{obs} = \frac{\hat{\theta}}{\hat{\sigma}\sqrt{\sum_{i=1}^{t}\frac{a_i^2}{r_i}}} \leftarrow SE(\hat{\theta}) \sim t_{\sum r_i - t}$

Anova for unbalance CRD

Source	SS	df.	MS	F
trt	$\sum_{i=1}^{t} r_i(\bar{Y}_{i\cdot} - \bar{Y}_{\cdot\cdot})^2$	$/\ t-1$	$= \hat{\sigma}_t^2$	F_{obs}
Residual	$\sum_{i=1}^{t}\sum_{j=1}^{r_i}(\bar{Y}_{ij} - \bar{Y}_{i\cdot})^2$	$/\ \sum_{i=1}^{t} r_i - t$	$= \hat{\sigma}^2$	
total	$\sum_{i=1}^{t}\sum_{j=1}^{r_i}(\bar{Y}_{ij} - \bar{Y}_{\cdot\cdot})^2$	$\sum_{i=1}^{t} r_i - 1$		

↳ TSS = $\sum r_i - 1$ · overall Var.

$P(F_{t-1, \sum r_i - t} \geq c) = \alpha$

reject $H_0 \Rightarrow F_{obs} > C$

Randomize block design (CRD w/ blocking)

$Y_{ij} = \mu + \tau_i + \beta_j + \epsilon_{ij}$

↳ response in treatment i, blocking j

$i = 1 \ldots t$, $j = 1 \ldots b$

$\hat{\mu} = \bar{Y}_{..}$ $\quad$ $df = (t-1)(b-1)$

$\hat{\tau}_i = \bar{Y}_{i} - \bar{Y}_{..}$

$\beta_j = \bar{y}_{.j} - \bar{y}_{..}$ $\quad$ (assume same trt effect in block)

$$\hat{\sigma}^2 = \frac{1}{(t-1)(b-1)} \sum_{i=1}^{t} \sum_{j=1}^{b} (y_{ij} - \hat{\mu} - \hat{\tau}_i - \hat{\beta}_j)^2$$

↳ $y_{ij} - \bar{y}_{i.} - \bar{y}_{.j} + \bar{y}_{..}$

$\Rightarrow$ TSS = trtSS + RSS (w/ blocking) + Block SS

$Y_{ij} \sim N(\mu + \tau_i + \beta_j, \sigma^2)$

$\bar{Y}_{i.} \sim N(\mu + \tau_i, \frac{\sigma^2}{b})$

$\hat{\theta} = \sum_{i=1}^{t} a_i \hat{\tau}_i = \sum_{i=1}^{t} a_i \bar{Y}_{i.} \sim N(\sum a_i \tau_i, \sigma^2 \frac{\sum a_i^2}{b})$

Anova

Source	SS		df		MS	F
trt	$b\sum_{i=1}^{t}(\bar{Y}_{i.} - \bar{Y}_{..})^2$	/	$t-1$	=	$\hat{\sigma}_t^2$	$F = \frac{\hat{\sigma}_t^2}{\hat{\sigma}^2}$
Block	$t\sum_{j=1}^{b}(\bar{Y}_{.j} - \bar{Y}_{..})^2$	/	$b-1$	=	$\hat{\sigma}_b^2$	
Residual	$\sum_i^t \sum_j^b (Y_{ij} - \bar{Y}_{i.} - \bar{Y}_{.j} - \bar{Y}_{..})^2$	/	$(t-1)(b-1)$	=	$\hat{\sigma}^2$	
total	$\sum_i^t \sum_j^b (Y_{ij} - \bar{Y}_{..})^2$		$tb-1$			

Interaction (one effect depend on another?)

Factorial design (w/ 2 factor)

Factor A,B w/ t_a, t_b levels

trt combination = AB

If effect of A is same regardless level B and vice versa, $\Rightarrow$ effects are additive.

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$$

α_i, β_j = effect of factor A, B

$(\alpha\beta)_{ij}$ = Interaction effect $= \gamma_{ij}$

$$\hat{\mu} = \bar{Y}_{...}$$

$$\hat{\alpha}_i = \bar{Y}_{i..} - \bar{Y}_{...}, \quad \hat{\beta}_i = \bar{Y}_{.j.} - \bar{Y}_{...}$$

$$\hat{\gamma}_{ij} = \bar{Y}_{ij.} - \bar{Y}_{i..} - \bar{Y}_{.j.} + \bar{Y}_{...}$$

$$\hat{\sigma}^2 = \frac{1}{t_a t_b (r-1)} \underbrace{\sum_{i=1}^{t_a} \sum_{j=1}^{t_b} \sum_{k=1}^{r} (Y_{ijk} - \bar{Y}_{ij.})^2}_{RSS}$$

$$SST = \underbrace{r \sum_i^{t_a} (\bar{Y}_{i..} - \bar{Y}_{...})^2}_{SSA} + \underbrace{r \sum_j^{t_b} (\bar{Y}_{.j.} - \bar{Y}_{...})^2}_{SSB}$$

$$+ \underbrace{r \sum_i^{t_a} \sum_j^{t_b} (\bar{Y}_{ij.} - \bar{Y}_{i..} - \bar{Y}_{.j.} + \bar{Y}_{...})^2}_{SSI}$$

Factorial ANOVA w/ Interaction (A & B)

Source	SS		df		MS	F
A	SSA	/	$t_a - 1$	=	MSA	$MSA/\hat{\sigma}^2$
B	SSB	/	$t_b - 1$	=	MSB	$MSB/\hat{\sigma}^2$
Interaction	SSI	/	$(t_a-1)(t_b-1)$	=	MSI	$MSI/\hat{\sigma}^2$
Residual	RSS	/	$t_a t_b (r-1)$	=	$\hat{\sigma}^2$	
Total	TSS					

Two Factor w/ blocking

t trt, b blocks, 1 replicate per trt/block combo

total obs = tb

RBD in terms of additive & Interaction ($t = t_a t_b$)

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \lambda_k + \epsilon_{ij}$$

λ_k = block effect, $i = 1 \dots t_a$, $j = 1 \dots t_b$

$k = 1 \dots b$ (blocks)

$\hat{\mu} = \bar{Y}_{\cdots}$, $\hat{\lambda}_k = \bar{Y}_{\cdot\cdot k} - \bar{Y}_{\cdots}$

$\alpha_i = \bar{Y}_{i\cdot\cdot} - \bar{Y}_{\cdots}$, $\beta_j = \bar{Y}_{\cdot j \cdot} - \bar{Y}_{\cdots}$

$\gamma_{ij} = \bar{Y}_{ij\cdot} - \bar{Y}_{i\cdot\cdot} - \bar{Y}_{\cdot j\cdot} + \bar{Y}_{\cdots}$

$$\hat{\sigma}^2 = \frac{1}{(t_a t_a - 1)(b-1)} \sum_i^{t_a} \sum_j^{t_b} \sum_k^{b} \underbrace{(\hat{Y}_{ijk} - \bar{Y}_{ij\cdot} - \bar{Y}_{\cdot\cdot k} + \bar{Y}_{\cdots})}_{RSS}$$

$TSS = b \sum_{ij} (\bar{Y}_{ij} - \bar{Y}_{\cdots})$,

$\text{block SS} = t_a t_b \sum_k (\bar{Y}_{\cdot\cdot k} - \bar{Y}_{\cdots})^2$

Factorial ANOVA w/ Interaction (A & B)

Source	SS		df		MS	F
A	SSA	/	$t_a - 1$	=	MSA	$MSA/\hat{\sigma}^2$
B	SSB	/	$t_b - 1$	=	MSB	$MSB/\hat{\sigma}^2$
Interaction	SSI	/	$(t_a-1)(t_b-1)$	=	MSI	$MSI/\hat{\sigma}^2$
Residual	RSS	/	$b t_a t_b - 1$	=	$\hat{\sigma}^2$	
Block	BSS	/	$b-1$	=	BMS	

Workflow.

diff among trts? $\xrightarrow{\text{No}}$ done

(seperate trt. ANOVA)

$\downarrow$ Yes

Significant Interactions? $\xrightarrow{\text{No}}$ check A, B, significant individually.

($MSI/\hat{\sigma}^2$)

$\downarrow$ Yes

Done visualize w/ interaction plot